Hypothermia / Frostbite

TE-05 | Toxicology & Environmental

Quick flow

01 Remove Wet Clothing [EMT / AEMT / Paramedic]

Move to warm shelter, dry patient, insulate, and measure temperature without delaying treatment.

02 Localized or Systemic? [EMT / AEMT / Paramedic]

Frostbite: no rubbing; if refreezing can be prevented, rapidly rewarm in 37-40°C (99-104°F) water for 20-30 minutes. Systemic: airway, glucose, warming, ECG.

03 Pulse Check ≥60 Seconds [EMT / AEMT / Paramedic]

If no pulse, begin the age-appropriate arrest pathway with hypothermia modifications and concurrent rewarming.

04 Active Core Warming [EMT / AEMT / Paramedic]

Warm blankets/packs to axillae and groin; protect skin and keep chemical heat away from frostbitten tissue; warmed fluids when available.

05 Support Perfusion [AEMT / Paramedic]

For hypotension or poor perfusion, give warmed isotonic crystalloid in reassessed aliquots: adult 500 mL; pediatric 20 mL/kg. Do not use a routine fixed maximum.

06 Hypothermic Arrest / Transport [EMT / AEMT / Paramedic]

Below 30°C (86°F): full resuscitation with rewarming. For VF/pVT, defibrillate once; if unsuccessful, defer further shocks and epinephrine until ≥30°C. Contact Medical Control early and consider ECLS-capable destination when feasible.

Overview

- Remove wet clothing, protect from further heat loss, and handle severe hypothermia carefully while assessing for signs of life for at least 60 seconds.
- Treat localized cold injury without rubbing, massage, or allowing refreezing.

Provider-Level Actions

Protect rescuers, stop exposure when safe, support ABCs, and involve Poison Control or Medical Control early.

EMT

- Remove wet clothing, dry/insulate, apply protected heat packs to axillae/groin, check glucose, and provide gentle handling.

AEMT

- Perform all EMT care plus IV/IO access and warmed isotonic crystalloid when available for hypotension or poor perfusion, with reassessment after each aliquot.

Paramedic

- Perform all prior care plus cardiac monitoring and the hypothermic-arrest sequence below 30°C (86°F): full resuscitation with rewarming; one VF/pVT defibrillation attempt, then defer further shocks and epinephrine until ≥30°C.

Indications

- Systemic hypothermia or localized cold injury/frostbite.

Medications

Warmed isotonic crystalloid

Adult 500 mL IV/IO aliquots; pediatric 20 mL/kg IV/IO aliquots; repeat only after reassessment for hypotension or poor perfusion.

Warnings / precautions

- Do not rub or massage frostbitten tissue, debride frostbite blisters, or apply direct chemical heat to frostbitten tissue.
- Do not rewarm frostbite if refreezing is possible. When sustained thaw can be assured, use 37-40°C (99-104°F) water for 20-30 minutes.
- Below 30°C (86°F), provide full resuscitation with rewarming. For VF/pVT, defibrillate once; if unsuccessful, defer further shocks and epinephrine until ≥30°C.

Clinical pearls

- Do not use a fixed temperature alone to stop resuscitation. Continue unless there is obvious lethal injury or the patient is frozen solid; seek early Medical Control and consider ECLS-capable destination when feasible.

Source and references

Claiborne County EMS approved protocol manual. | Poison Help national line: 1-800-222-1222. | Current Tennessee EMS scope of practice and Claiborne County standing orders control for all provider permissions.

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